# Knowledge index: team questions and open inputs

A documented answer does not certify the target hero recipe. Conditional questions expose the missing data, measurements or access.

## What model and token horizon can96H100GPUs support in30days?

Contract `hero-compute` · status **conditional**.

- Compute, parameters and a constrained hero budget

Evidence: `claim-budget-96x30`, `claim-looped-compute-accounting`.

Remaining inputs: Need GPU SKU/topology, measured useful throughput, qualified trainer/recurrence and serving constraints.

## How do unique retained tokens, repeated exposures and phase weights determine feasible data demand?

Contract `data-supply` · status **conditional**.

- Data-mixture experiments: supply, exposure, controls and transfer

Evidence: `claim-marin-phase-cap`, `claim-harrier-luxical-separation`.

Remaining inputs: Need accessible prepared sources and a real retained-token/loader census under the chosen tokenizer.

## How do we reproduce Marin topic/quality buckets rather than only its search records?

Contract `bucket-reproduction` · status **conditional**.

- Marin replication: a stage-by-stage evidence and acceptance contract

Evidence: `claim-marin-normalized-id`, `claim-marin-quality-compatibility`, `claim-marin-quality-alignment`.

Remaining inputs: Need historical compatible embeddings/scorer/calibration/store access, or declare a newly rebuilt taxonomy.

## What does width-transfer evidence establish about width, depth and expert choices?

Contract `width-depth` · status **conditional**.

- Width, batch and horizon: a conditional hyperparameter-transfer protocol
- Compute, parameters and a constrained hero budget

Evidence: `claim-mup-scope`, `claim-steplaw-widths`.

Remaining inputs: Need matched shape experiments and target-hardware throughput; no optimal depth/expert count is established for our target.

## How should we tune LR, token batch, decay and schedule before fitting transfer laws?

Contract `optimizer-search` · status **conditional**.

- Width, batch and horizon: a conditional hyperparameter-transfer protocol

Evidence: `claim-steplaw-law`, `claim-power-lines-timescale`.

Remaining inputs: Need resolved group conventions and actual proxy grid/repeat/holdout training observations.

## What experiment grid makes size/horizon laws identifiable and tests extrapolation?

Contract `fit-identifiability` · status **documented**.

- Identifiable scaling fits and independent larger holdouts

Evidence: `claim-fixed-tpp-identifiability`.

## Which evaluation contracts make checkpoints and training stages comparable?

Contract `evaluation` · status **documented**.

- Evaluation contracts: losses, capabilities, contexts and failures

Evidence: `claim-evaluation-denominators`.

## Which Marin long-context experiments were configured/completed and what winner is actually established?

Contract `long-context` · status **conditional**.

- Marin replication: a stage-by-stage evidence and acceptance contract
- Evaluation contracts: losses, capabilities, contexts and failures

Evidence: `claim-long-context-treatment`, `claim-long-context-completions`.

Remaining inputs: Need complete paired per-length final capability outcomes; captured8xlaunch alone supplies no winning-skew conclusion.

## What SFT/RL evidence establishes systems readiness versus broad capability gains?

Contract `posttraining` · status **conditional**.

- Marin replication: a stage-by-stage evidence and acceptance contract
- Diagnosing a failed scaling or training decision

Evidence: `claim-sft-smoke-boundary`, `claim-rsi-tradeoff`.

Remaining inputs: Need complete post-training runs and independent capability/guardrail evaluations for the chosen target.

## How do we add high-quality cyber data without losing attribution or generalization checks?

Contract `cyber-extension` · status **open**.

- Data-mixture experiments: supply, exposure, controls and transfer
- Evaluation contracts: losses, capabilities, contexts and failures
- Marin replication: a stage-by-stage evidence and acceptance contract

Evidence: `claim-marin-transfer-d1536`, `claim-rsi-tradeoff`.

Remaining inputs: Need selected cyber sources, quality/contamination fixtures, held-out tasks/families and matched training controls.